

SaturnCFD: A Distributed Unstructured Finite-Volume Solver

Benchmark Report

SaturnCFD submission

August 4, 2026

Abstract

SaturnCFD is an original C++17/MPI cell-centred unstructured solver for the two-dimensional compressible Euler and laminar Navier–Stokes equations. It combines CGNS multi-zone topology, METIS graph partitioning, neighbour-only halos, limited second-order reconstruction, a Rusanov flux, Newton–Fourier viscous terms, 4-by-4 block implicit defect correction, and frozen-history BDF2. The run-status table identifies the actual completion status for every required case; every numerical claim and figure in this report is regenerated from the submitted CSV and parallel VTK files.

1 Governing equations and nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and the solved system is

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}^i + \partial_y \mathbf{G}^i = \partial_x \mathbf{F}^v + \partial_y \mathbf{G}^v.$$

The inviscid fluxes are

$$\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, u(\rho E + p)]^T, \quad \mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, v(\rho E + p)]^T.$$

A calorically perfect gas closes the system: $p = (\gamma - 1)\rho e$, $E = e + (u^2 + v^2)/2$, and $a = \sqrt{\gamma p/\rho}$, with $\gamma = 1.4$ and $R = 1$. Reference values are $\rho_\infty = U_\infty = L = A = 1$, p is scaled by $\rho_\infty U_\infty^2$, and force coefficients use $q_\infty A$ with $q_\infty = \rho_\infty U_\infty^2/2$. For laminar cases, $\mu = \rho_\infty U_\infty L Re/Re$, $\tau_{xx} = 2\mu u_x - 2\mu(u_x + v_y)/3$, $\tau_{yy} = 2\mu v_y - 2\mu(u_x + v_y)/3$, $\tau_{xy} = \mu(u_y + v_x)$, and $\mathbf{q} = -k\nabla T$ with $k = \mu c_p/Pr$ and $Pr = 0.72$.

2 Meshes, topology, and MPI decomposition

The reader imports every unstructured zone and TRI/QUAD volume section. CGNS donor PointLists union abutting vertices before a canonical edge map is formed; hence the five cylinder seam sections become interior faces rather than artificial walls. NACA has 15,682 points, 20,816 cells and 484 physical boundary faces. The two-zone cylinder has 10,185 cells and 120 physical boundary faces after 320 duplicate interface vertices are merged. Case-file family mappings, not mesh-name branches, select farfield, slip, and no-slip adiabatic conditions.

Case	mesh	cells	M_∞	Re	step/ t_f	CFL; inner
naca0012.m015.inviscid	NACA0012.H2.cgns	20816	0.15	–	20000	1→100; 3–50
naca0012.m080.inviscid	NACA0012.H2.cgns	20816	0.8	–	30000	1→100; 3–50
naca0012.m200.inviscid	NACA0012.H2.cgns	20816	2	–	40000	0.2→50; 3–80
naca0012.m015.laminar_re5000	NACA0012.H2.cgns	20816	0.15	5000.0	40000	1→100; 3–80
naca0012.m080.laminar_re5000	NACA0012.H2.cgns	20816	0.8	5000.0	40000	0.5→80; 3–80
naca0012.m200.laminar_re5000	NACA0012.H2.cgns	20816	2	5000.0	50000	0.2→50; 3–100
cylinder.m010.laminar_re20	CylinderB1.cgns	10185	0.1	20.0	30000	1→100; 3–80
cylinder.m010.laminar_re200	CylinderB1.cgns	10185	0.1	200.0	300.0	1→1; 5–1000

METIS partitions the cell adjacency graph. Rank zero performs this serial preprocessing once, sends each rank only its owned cells, one-ring ghost cells, incident faces, and static neighbour lists, then releases the global mesh and all other partitions. Iterations exchange only packed state or gradient/limiter data with neighbour-scoped `MPI_Irecv/Isend`; reductions are used for norms and forces. The submitted `partition_diagnostics.csv` files provide the measured owned/ghost/send/receive counts and METIS edge cuts.

Case	rank	owned	ghost	neighbours	send	receive	edge cut
naca0012.m015_inviscid	0	2569	51	1	51	51	388
naca0012.m015_inviscid	1	2574	184	7	184	184	388
naca0012.m015_inviscid	2	2640	102	3	103	102	388
naca0012.m015_inviscid	3	2635	90	3	90	90	388
naca0012.m015_inviscid	4	2603	82	3	82	82	388
naca0012.m015_inviscid	5	2626	86	3	85	86	388
naca0012.m015_inviscid	6	2589	93	3	93	93	388
naca0012.m015_inviscid	7	2580	83	3	83	83	388
cylinder.m010_laminar_re20	0	1269	123	6	126	123	522
cylinder.m010_laminar_re20	1	1275	93	3	94	93	522
cylinder.m010_laminar_re20	2	1265	113	3	115	113	522
cylinder.m010_laminar_re20	3	1266	148	6	140	148	522
cylinder.m010_laminar_re20	4	1274	133	4	134	133	522
cylinder.m010_laminar_re20	5	1271	121	4	123	121	522
cylinder.m010_laminar_re20	6	1284	119	4	121	119	522
cylinder.m010_laminar_re20	7	1281	118	4	115	118	522

3 Spatial discretization and boundaries

For cell i , $V_i \dot{\mathbf{U}}_i + \sum_f (\hat{\mathbf{F}}^i - \hat{\mathbf{F}}^v)_f S_f = 0$. Primitive gradients use inverse-distance weighted least squares with conditioning regularization. Piecewise-linear face values are limited componentwise by Barth–Jespersen bounds; density and pressure receive an additional positive-face scaling and conservative updates use a common positivity line search. Reconstruction is active for every production result. If a reconstructed face has non-finite density/pressure or a value at/below its positivity floor, that face alone falls back to its adjacent cell-centre primitive state; if a conservative update remains inadmissible, its line search reduces the update. The measured counts below identify every affected submitted case. Steady cases use a conservative Rusanov/LLF numerical flux with unit dissipation scale. The transient cylinder uses HLLC to retain contact/shear waves, with unit-scale Rusanov as a positivity fallback. For the transient HLLC path, the fallback is taken for a non-finite/degenerate star-state denominator, non-positive/non-finite star pressure, inadmissible star state, or non-finite flux; steady runs intentionally use Rusanov, so their final column is an evaluation count rather than a failure count. Farfield states use incoming freestream and outgoing interior Riemann invariants, which avoids reflective low-Mach pseudo-time modes. Slip walls use zero normal mass flux and direct pressure momentum flux. No-slip walls impose $u = v = 0$ and correct wall-normal velocity gradients; the pressure gradient is corrected to enforce zero normal temperature gradient. Interior viscous gradients are arithmetic averages plus a two-point normal correction. Reported viscous force is tangential skin friction; normal viscous traction is not mislabeled as friction. Surface CSV velocities are boundary values rather than adjacent cell-centre values.

Case	reconstructed-state fallbacks	damped updates	Rusanov evaluations/fallbacks
naca0012.m015_inviscid	0	0	93612812
naca0012.m080_inviscid	0	0	262050206
naca0012.m200_inviscid	0	0	171939666
naca0012.m015_laminar_re5000	0	0	179418476
naca0012.m080_laminar_re5000	0	0	557517924
naca0012.m200_laminar_re5000	0	0	968414690
cylinder.m010_laminar_re20	0	0	79235292
cylinder.m010_laminar_re200	0	0	0

4 Implicit steady and transient integration

Steady cases use the supplied geometric CFL ramps and local spectral pseudo-time steps. Each defect correction assembles normal 4-by-4 Euler Jacobian blocks plus viscous spectral coupling; distributed block-Jacobi iterations solve for a conservative correction. This is an implicit multi-iteration path, not a diagonal explicit update. For the Reynolds-200 cylinder, BDF1 starts the calculation and BDF2 thereafter solves

$$\mathbf{G}(\mathbf{U}^{n+1}) = \frac{V}{2\Delta t} (3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}) + \mathbf{R}(\mathbf{U}^{n+1}) = 0.$$

The histories $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ remain frozen during all inner nonlinear/block iterations and shift only after acceptance. The production values are $\Delta t = 0.01$, $t_f = 300$, a minimum of five and maximum of 1000 nonlinear iterations, and a 10^{-3} reduction of the complete spatial-plus-BDF residual. Metadata reports actual min/mean/max iterations, misses, converged fraction, and the last ratio. The configured CFL of one caps globalization updates; the BDF mass-plus-spatial defect equation uses an inexact distributed 4-by-4 block-Jacobi correction (two to four linear sweeps per nonlinear update), without an additional pseudo-time diagonal. A deterministic localized cross-flow perturbation of amplitude $10^{-3}U_\infty$ breaks exact discrete symmetry; late-window statistics exclude the resulting startup transient.

5 Run status and force summary

Case	ranks	step	t_f	residual orders	status
naca0012_m015_inviscid	8	2566	0.00	5.046	converged
naca0012_m080_inviscid	8	7183	0.00	4.298	converged
naca0012_m200_inviscid	8	4713	0.00	3.775	converged
naca0012_m015_laminar_re5000	8	4918	0.00	5.577	converged
naca0012_m080_laminar_re5000	8	15282	0.00	4.294	converged
naca0012_m200_laminar_re5000	8	26545	0.00	4.381	converged
cylinder_m010_laminar_re20	8	3846	0.00	6.181	converged
cylinder_m010_laminar_re200	4	30000	300.00	3.026	statistically_periodic

Case	C_D	C_L	C_m	$C_{D,p}$	$C_{D,v}$
naca0012_m015_inviscid	0.00055512	8.18674e-05	7.12339e-05	0.00055512	0
naca0012_m080_inviscid	0.00815749	0.00157993	0.000258413	0.00815749	0
naca0012_m200_inviscid	0.0921625	0.000557498	8.5415e-05	0.0921625	0
naca0012_m015_laminar_re5000	0.0638984	0.000665257	-0.000149848	0.0194915	0.0444069
naca0012_m080_laminar_re5000	0.082115	0.000736706	7.98065e-05	0.0505453	0.0315697
naca0012_m200_laminar_re5000	0.117255	-0.000194598	-0.000129967	0.0787466	0.0385088
cylinder_m010_laminar_re20	2.03576	-5.95991e-05	-4.54669e-07	1.2293	0.806462
cylinder_m010_laminar_re200	1.26946	0.00635245	6.54483e-05	1.02706	0.242407

6 Computed results

6.1 naca0012_m015_inviscid

The submitted status is converged. The zero-incidence symmetry check gives terminal $C_L = 8.19e-05$. The submitted drag is $C_D = 0.0005551$, while the two wall branches provide the nontrivial pressure distribution shown below. The low-Mach rendering remains nearly symmetric and smoothly varying. Viscous force columns are identically zero and the wall velocity is the slip tangential projection. The first pair shows convergence and force evidence in fig. 1; fig. 2 renders Mach and pressure on the actual unstructured cells, and fig. 3 supplies the wall distribution.

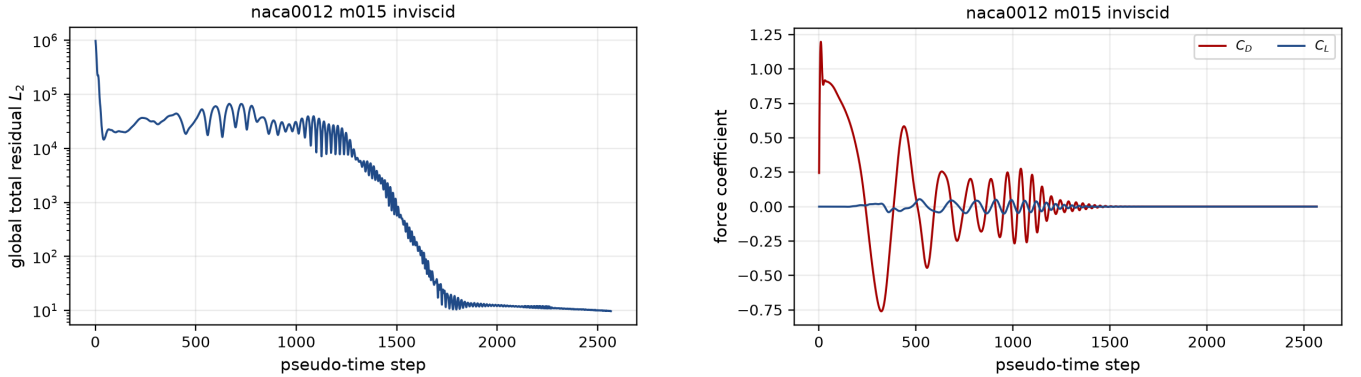


Figure 1: Residual and force histories for naca0012_m015_inviscid.

6.2 naca0012_m080_inviscid

The submitted status is converged. The zero-incidence symmetry check gives terminal $C_L = 0.00158$. The submitted drag is $C_D = 0.008157$, while the two wall branches provide the nontrivial pressure distribution shown below. The transonic rendering shows the strongest compressibility gradients near the body. Viscous force columns are identically zero and the wall velocity is the slip tangential projection. The first pair shows convergence and force evidence in fig. 4; fig. 5 renders Mach and pressure on the actual unstructured cells, and fig. 6 supplies the wall distribution.

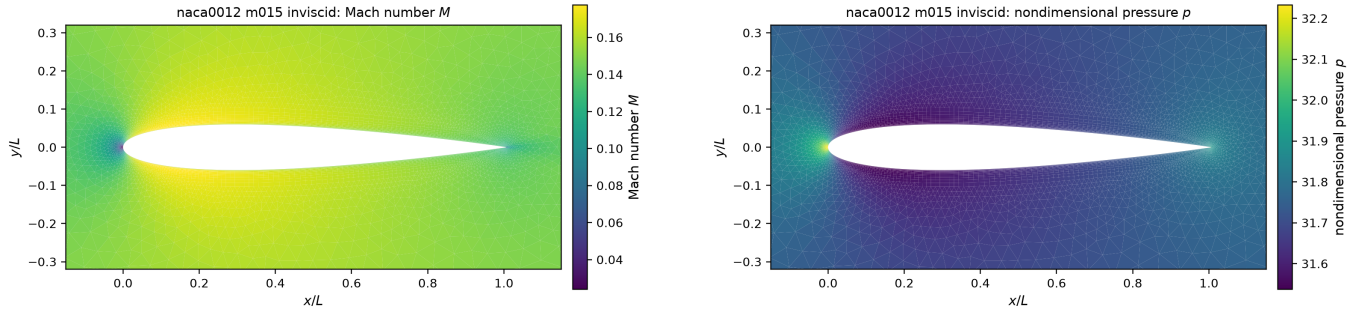


Figure 2: Computed Mach number and pressure for naca0012_m015_inviscid; colorbars name the plotted variables.

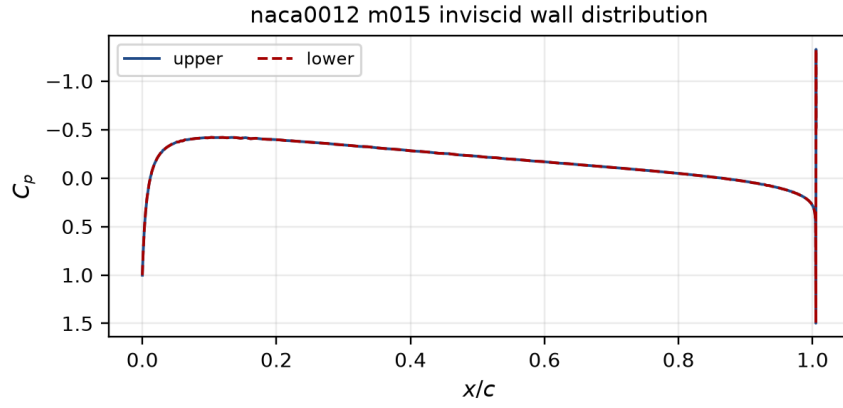


Figure 3: Wall pressure coefficient for naca0012_m015_inviscid.

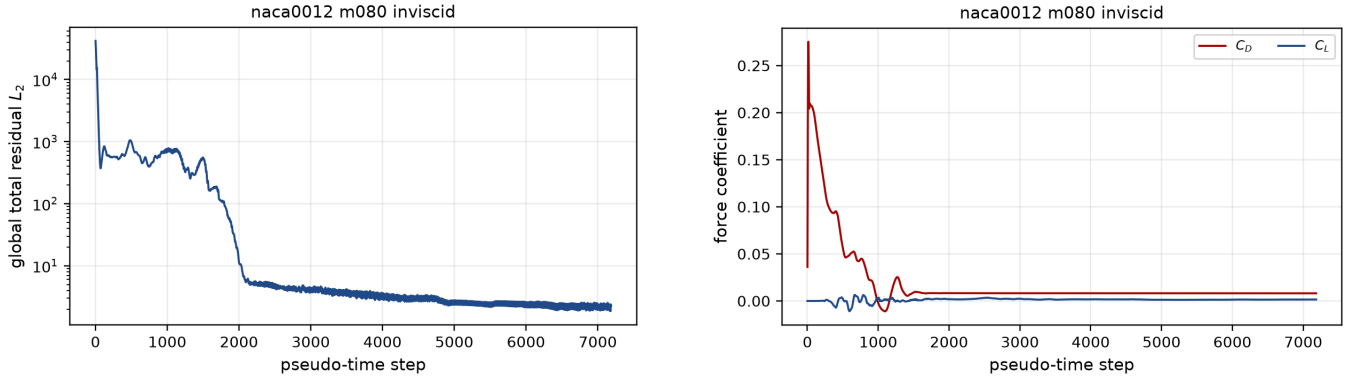


Figure 4: Residual and force histories for naca0012_m080_inviscid.

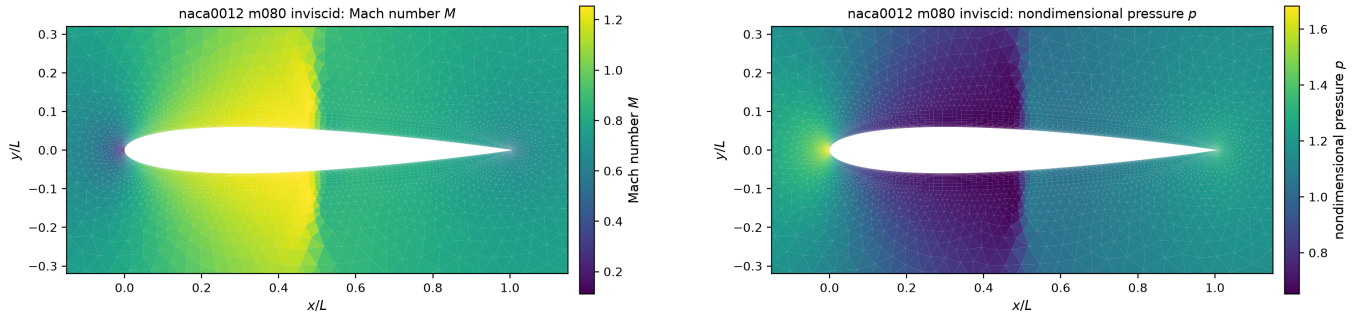


Figure 5: Computed Mach number and pressure for naca0012_m080_inviscid; colorbars name the plotted variables.

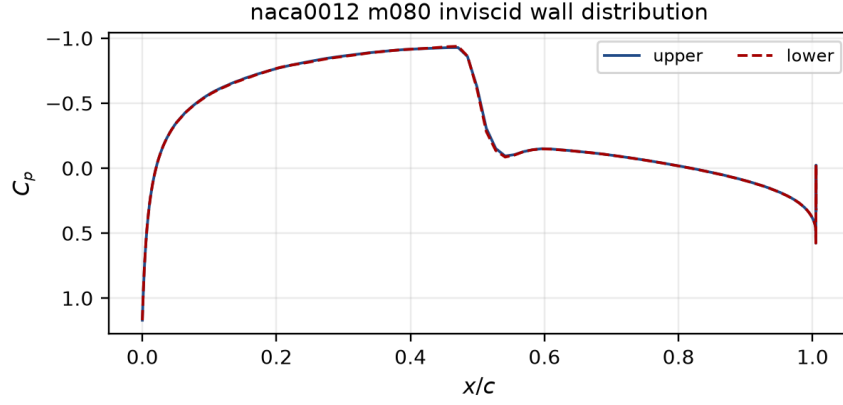


Figure 6: Wall pressure coefficient for naca0012_m080_inviscid.

6.3 naca0012_m200_inviscid

The submitted status is converged. The zero-incidence symmetry check gives terminal $C_L = 0.000557$. The submitted drag is $C_D = 0.09216$, while the two wall branches provide the nontrivial pressure distribution shown below. The pressure/Mach renderings resolve the leading-edge compression and supersonic shock pattern. Viscous force columns are identically zero and the wall velocity is the slip tangential projection. The first pair shows convergence and force evidence in fig. 7; fig. 8 renders Mach and pressure on the actual unstructured cells, and fig. 9 supplies the wall distribution.

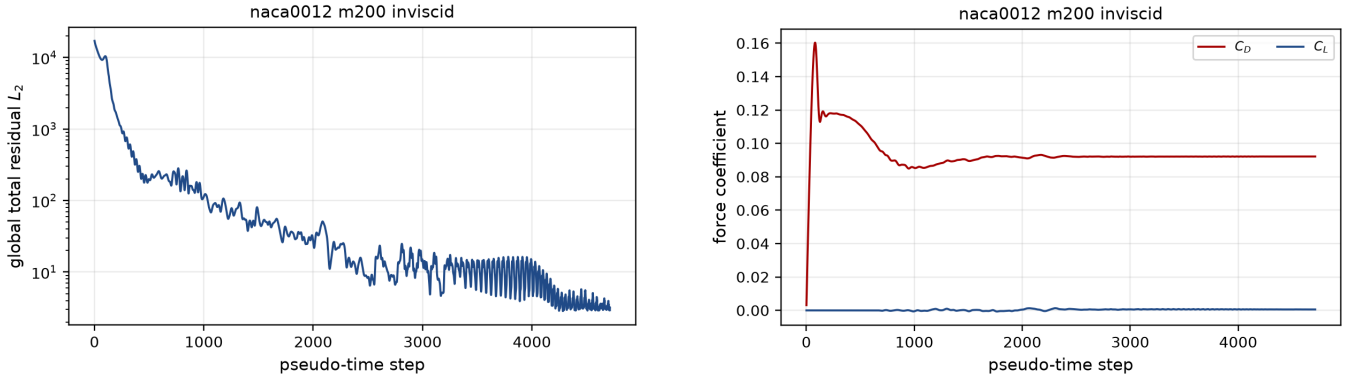


Figure 7: Residual and force histories for naca0012_m200_inviscid.

6.4 naca0012_m015_laminar_re5000

The submitted status is converged. The zero-incidence symmetry check gives terminal $C_L = 0.000665$. The submitted drag is $C_D = 0.0639$, while the two wall branches provide the nontrivial pressure distribution shown below. The low-Mach rendering remains nearly symmetric and smoothly varying. Tangential wall shear contributes $C_{D,v} = 0.04441$; the reported wall velocity is exactly the imposed no-slip boundary value. The first pair shows convergence and force evidence in fig. 10; fig. 11 renders Mach and pressure on the actual unstructured cells, and fig. 12 supplies the wall distribution.

6.5 naca0012_m080_laminar_re5000

The submitted status is converged. The zero-incidence symmetry check gives terminal $C_L = 0.000737$. The submitted drag is $C_D = 0.08212$, while the two wall branches provide the nontrivial pressure distribution shown below. The transonic rendering shows the strongest compressibility gradients near the body. Tangential wall shear contributes $C_{D,v} = 0.03157$; the reported wall velocity is exactly the imposed no-slip boundary value. The first pair shows

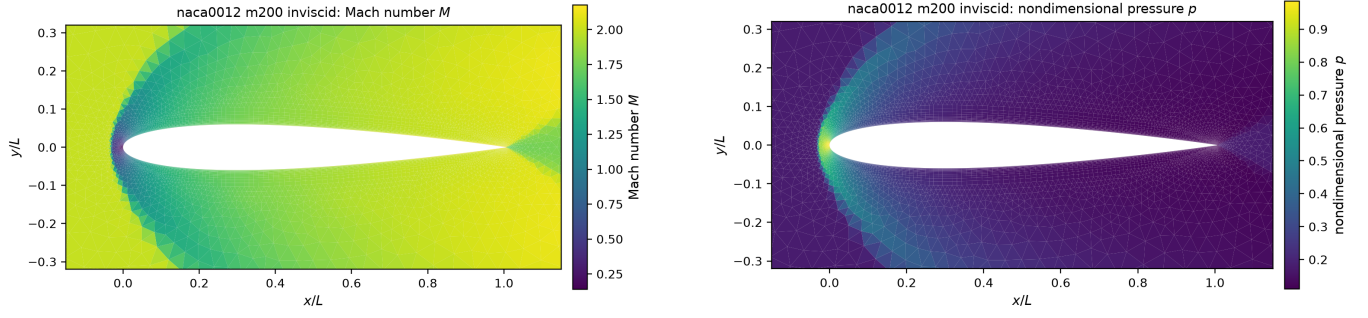


Figure 8: Computed Mach number and pressure for naca0012_m200_inviscid; colorbars name the plotted variables.

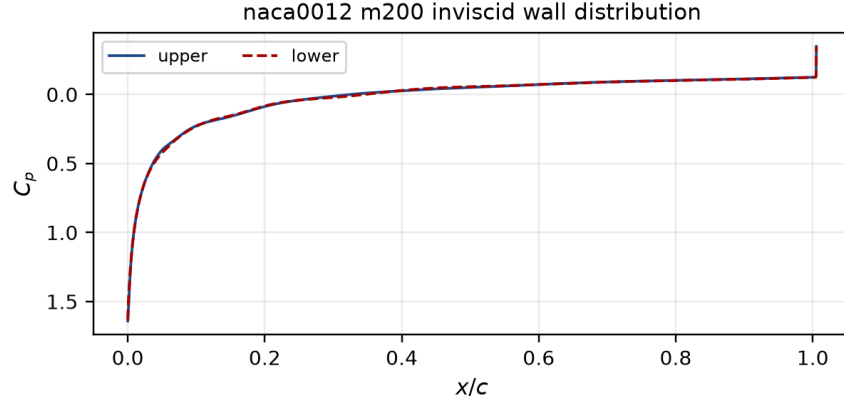


Figure 9: Wall pressure coefficient for naca0012_m200_inviscid.

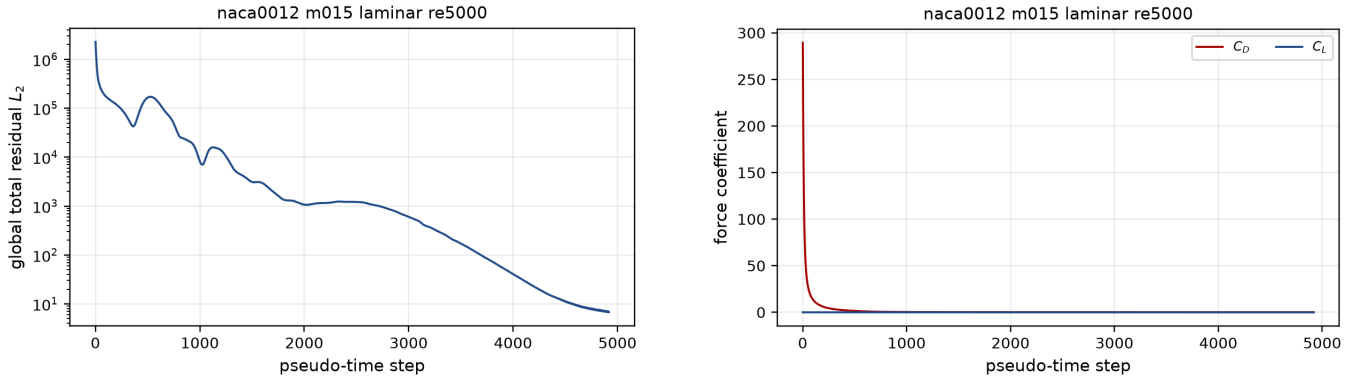


Figure 10: Residual and force histories for naca0012_m015_laminar_re5000.

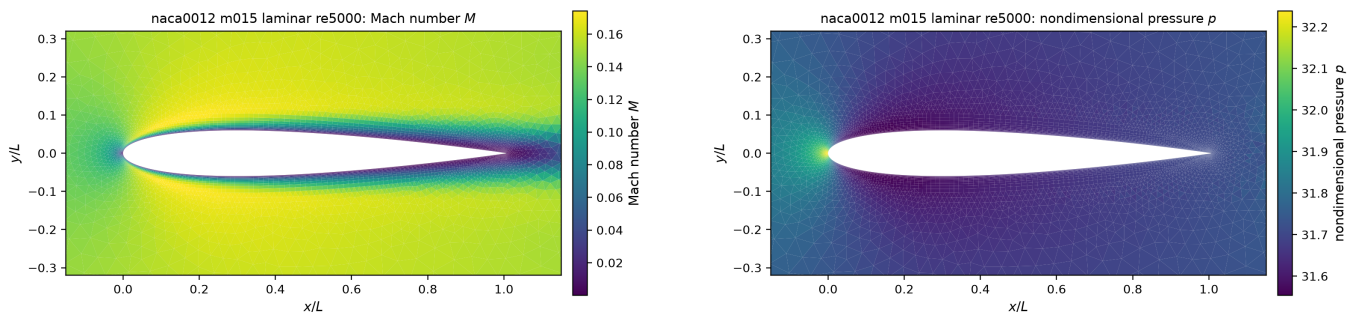


Figure 11: Computed Mach number and pressure for naca0012_m015_laminar_re5000; colorbars name the plotted variables.

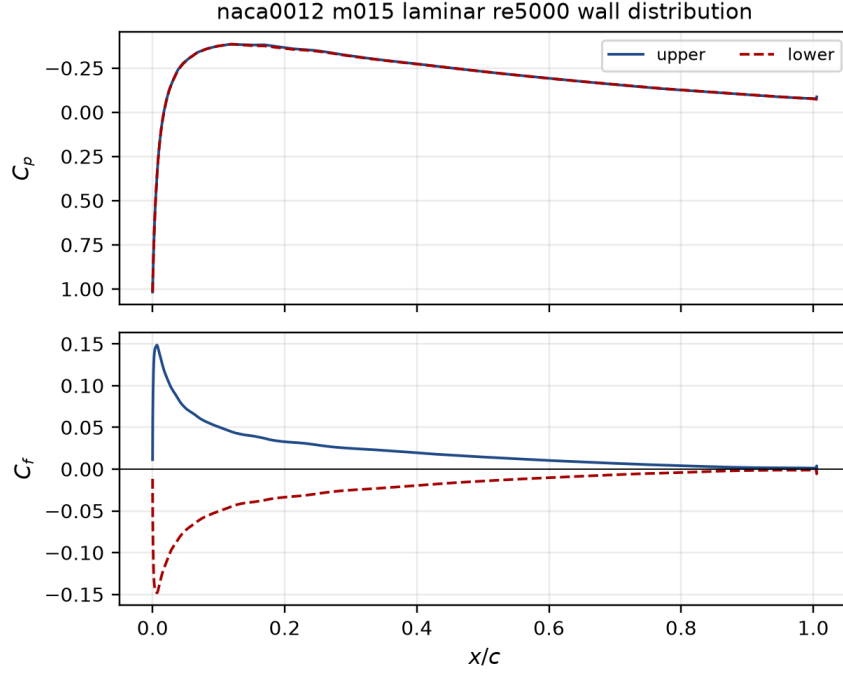


Figure 12: Wall pressure and tangential skin friction for naca0012_m015_laminar_re5000.

convergence and force evidence in fig. 13; fig. 14 renders Mach and pressure on the actual unstructured cells, and fig. 15 supplies the wall distribution.

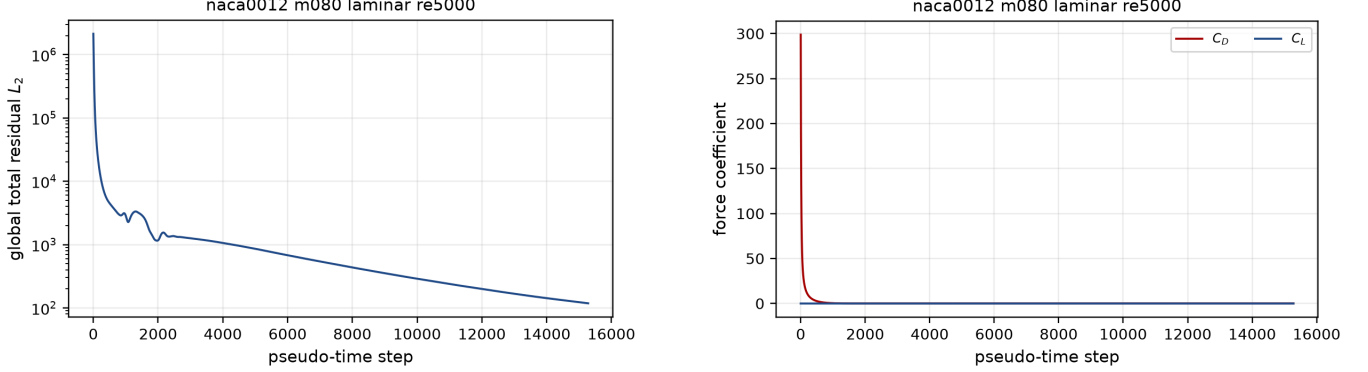


Figure 13: Residual and force histories for naca0012_m080_laminar_re5000.

6.6 naca0012_m200_laminar_re5000

The submitted status is converged. The zero-incidence symmetry check gives terminal $C_L = -0.000195$. The submitted drag is $C_D = 0.1173$, while the two wall branches provide the nontrivial pressure distribution shown below. The pressure/Mach renderings resolve the leading-edge compression and supersonic shock pattern. Tangential wall shear contributes $C_{D,v} = 0.03851$; the reported wall velocity is exactly the imposed no-slip boundary value. The first pair shows convergence and force evidence in fig. 16; fig. 17 renders Mach and pressure on the actual unstructured cells, and fig. 18 supplies the wall distribution.

6.7 cylinder_m010_laminar_re20

The submitted status is converged. The steady cylinder wake converged to positive drag $C_D = 2.036$ and near-zero mean lift $C_L = -5.96e-05$. The wake zoom and symmetric wall C_p/C_f curves are consistent with a steady laminar

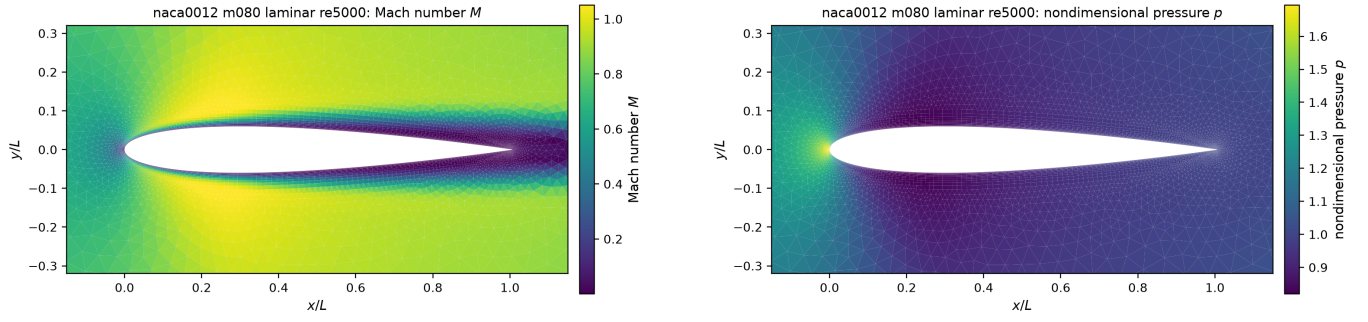


Figure 14: Computed Mach number and pressure for naca0012_m080_laminar_re5000; colorbars name the plotted variables.

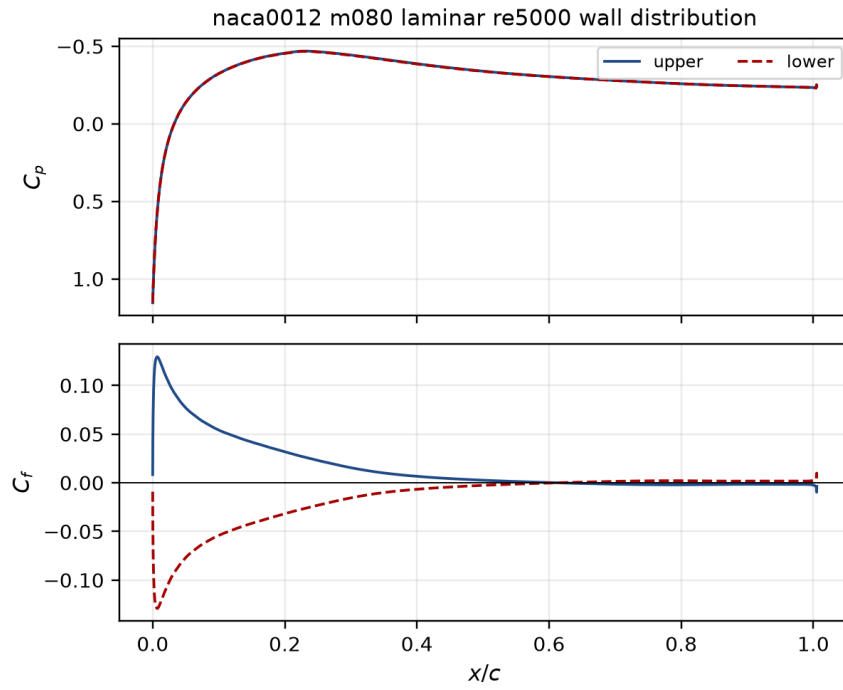


Figure 15: Wall pressure and tangential skin friction for naca0012_m080_laminar_re5000.

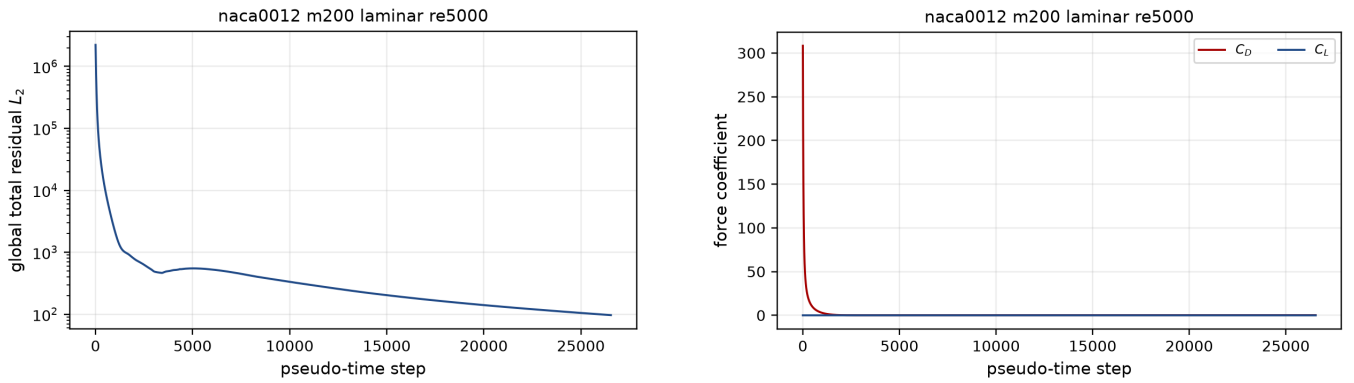


Figure 16: Residual and force histories for naca0012_m200_laminar_re5000.

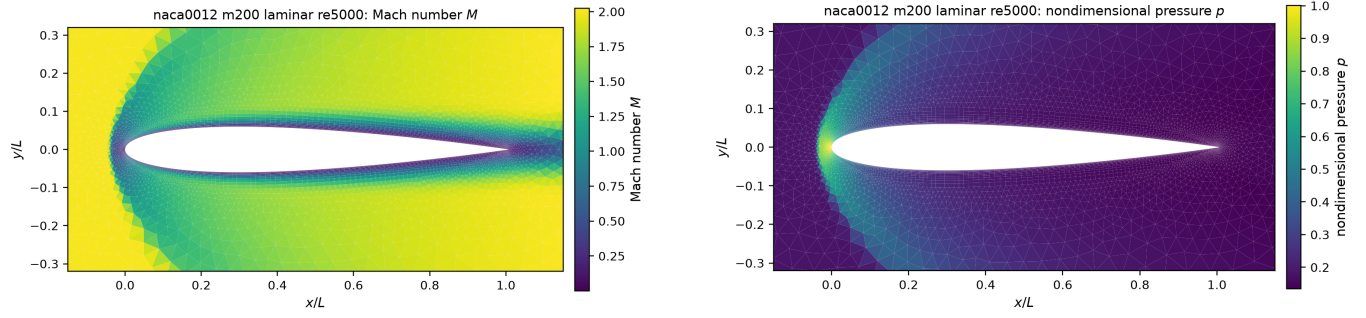


Figure 17: Computed Mach number and pressure for naca0012_m200_laminar_re5000; colorbars name the plotted variables.

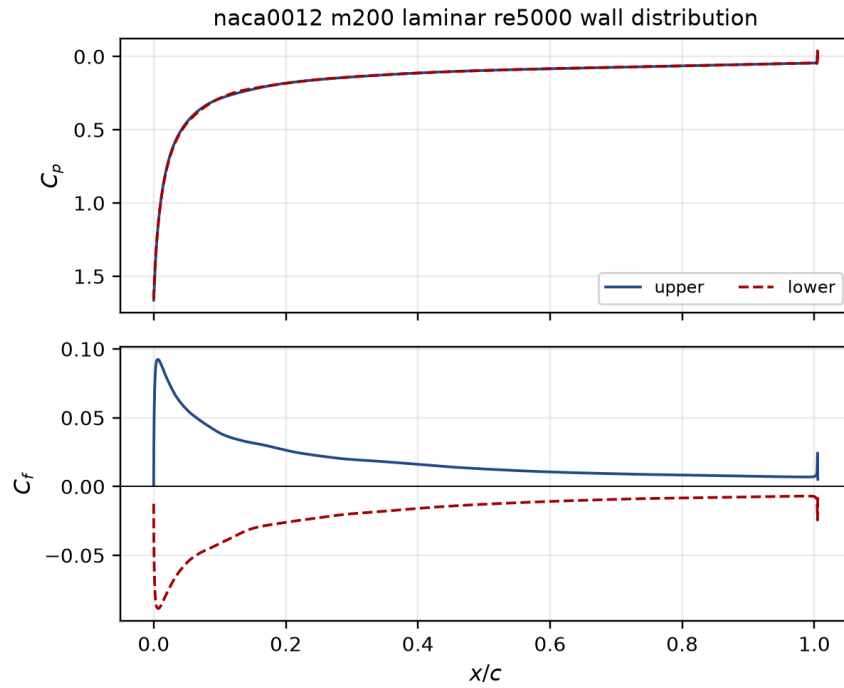


Figure 18: Wall pressure and tangential skin friction for naca0012_m200_laminar_re5000.

separated wake at Reynolds 20. The first pair shows convergence and force evidence in fig. 19; fig. 20 renders Mach and pressure on the actual unstructured cells, and fig. 21 supplies the wall distribution.

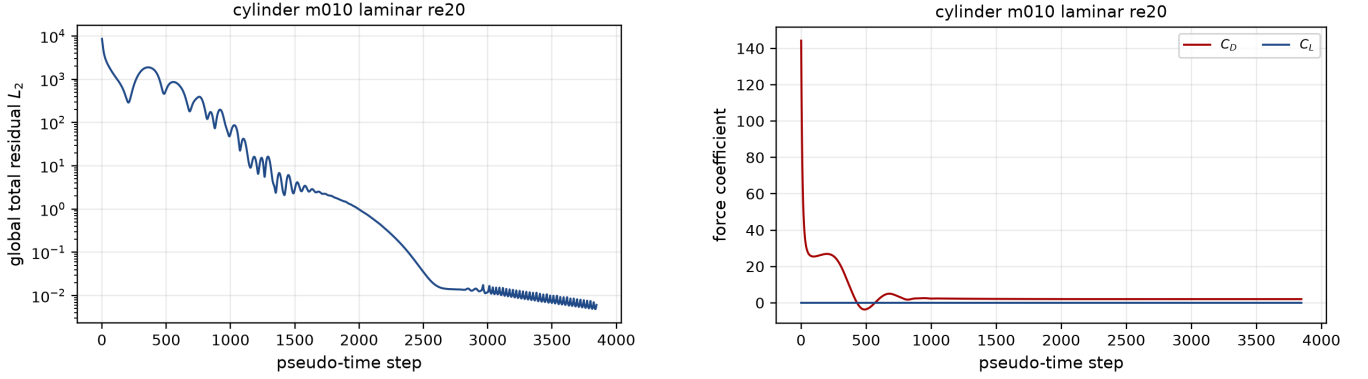


Figure 19: Residual and force histories for cylinder_m010_laminar_re20.

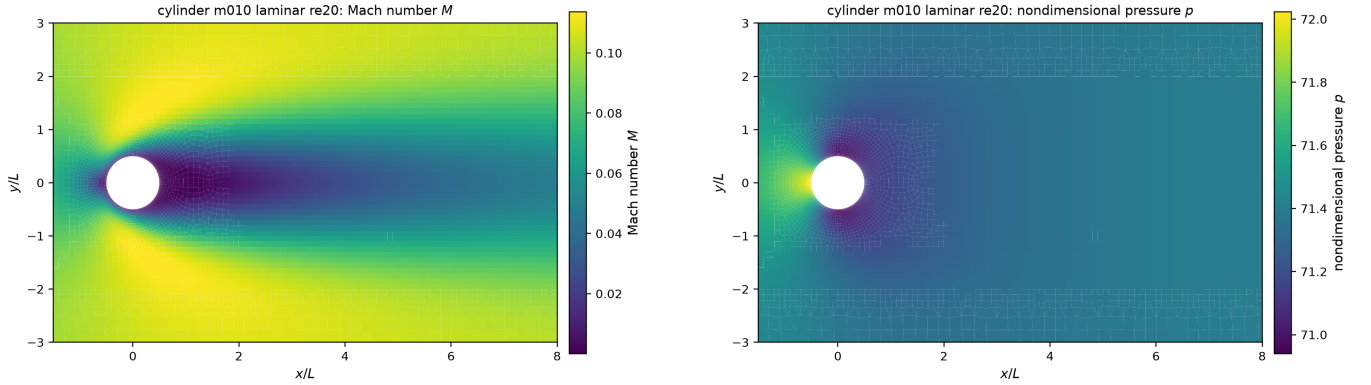


Figure 20: Computed Mach number and pressure for cylinder_m010_laminar_re20; colorbars name the plotted variables.

6.8 cylinder_m010_laminar_re200

The submitted status is statistically periodic. The full physical-time force history shows startup, growth of the antisymmetric mode, and a post-transient repeatable vortex-street regime. Quantitative late-window statistics and the clipped vorticity wake rendering are reported below. The first pair shows convergence and force evidence in fig. 22; fig. 23 renders Mach and pressure on the actual unstructured cells, and fig. 24 supplies the wall distribution. Over $t \geq 200.0$, the mean drag is 1.2695, lift RMS is 0.40067, robust lift amplitude is 0.56721, and the dominant frequency gives $St = fL/U_\infty = 0.18998$; the post-transient wake is shown in fig. 25.

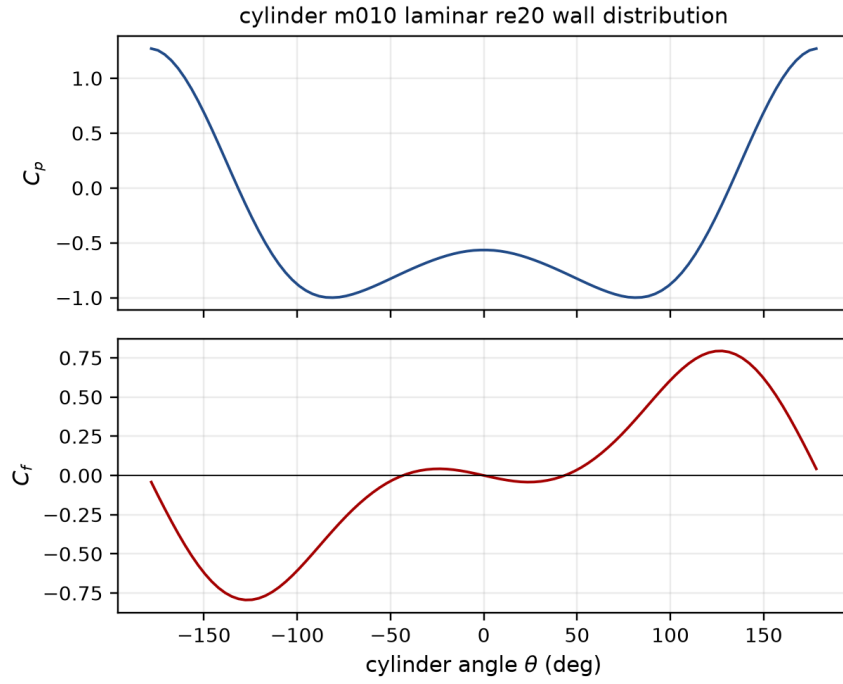


Figure 21: Wall pressure and tangential skin friction for cylinder_m010_laminar_re20.

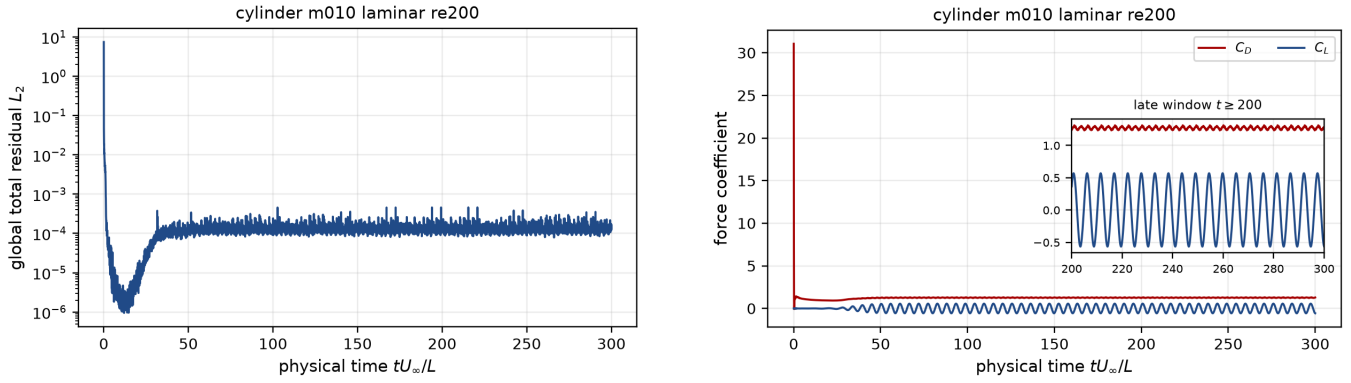


Figure 22: Residual and force histories for cylinder_m010_laminar_re200.

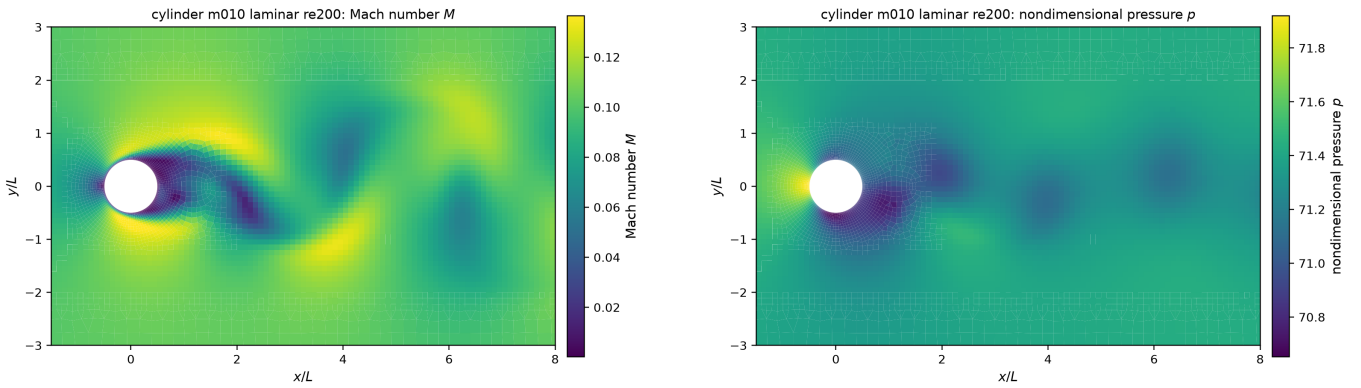


Figure 23: Computed Mach number and pressure for cylinder_m010_laminar_re200; colorbars name the plotted variables.

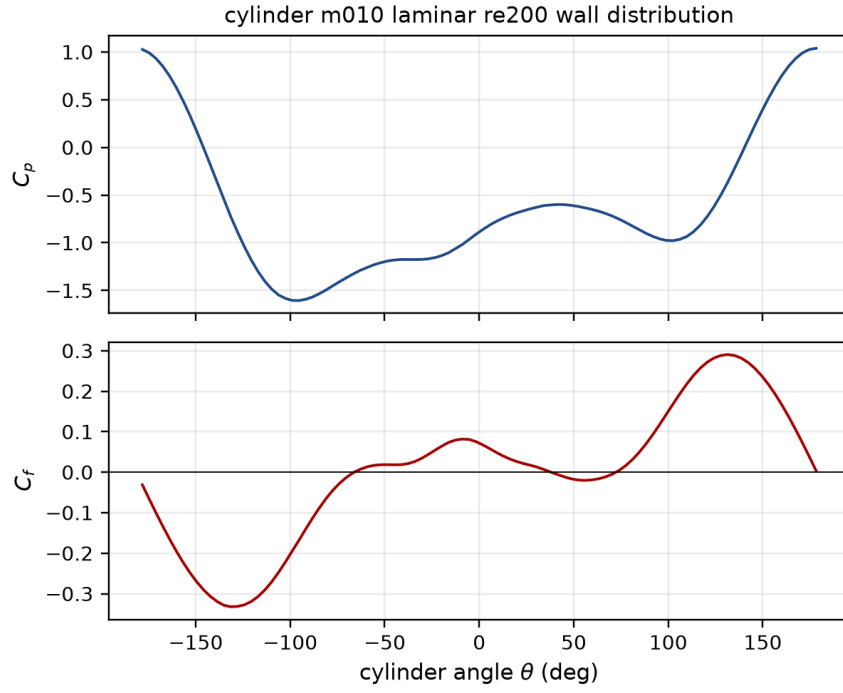


Figure 24: Wall pressure and tangential skin friction for cylinder_m010_laminar_re200.

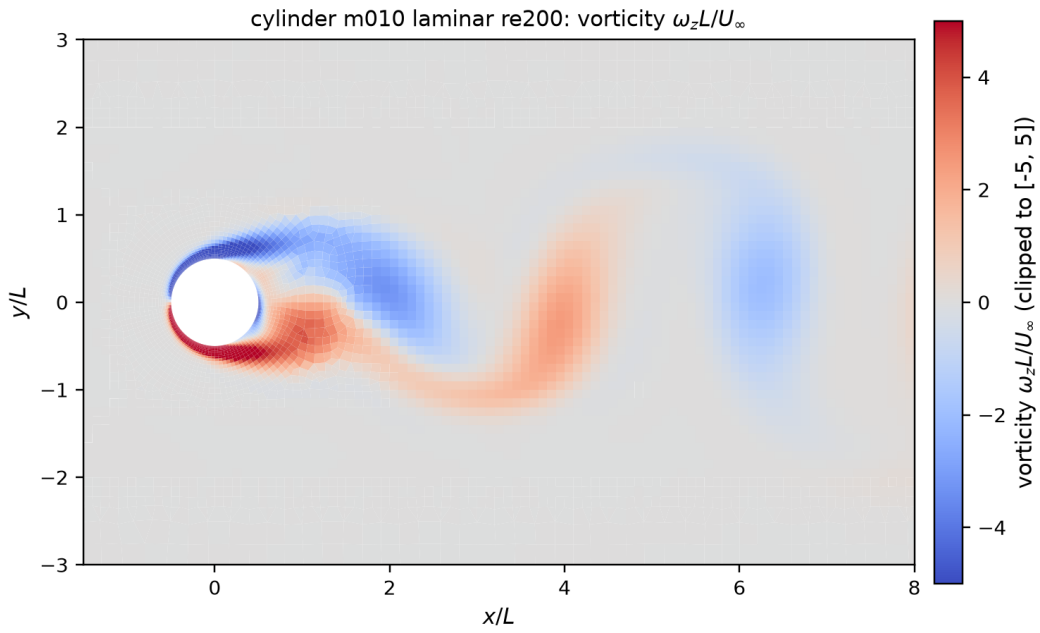


Figure 25: Post-transient cylinder wake vorticity, clipped to $[-5, 5]$ to preserve vortex-street contrast.

7 Parallel validation

The following independently generated comparisons use the actual MPI ranks recorded in `run.status.json`. The table records those ranks alongside final forces, terminal global residuals, timing, and the measured partition load balance.

Case	ranks	C_D	C_L	L_2 residual	load ratio	wall (s)	status
naca0012_m015_inviscid	1	0.0005551203	8.1867417e-05	9.7	1.0000	291.516	converged
naca0012_m015_inviscid	2	0.0005551203	8.1867417e-05	9.7	1.0006	186.531	converged
naca0012_m015_inviscid	8	0.0005551203	8.1867417e-05	9.7	1.0276	84.795	converged
cylinder_m010_laminar_re20	1	2.0357612	-5.959914e-05	0.00608	1.0000	301.752	converged
cylinder_m010_laminar_re20	2	2.0357612	-5.959914e-05	0.00608	1.0002	202.253	converged
cylinder_m010_laminar_re20	8	2.0357612	-5.959914e-05	0.00608	1.0150	75.564	converged

The accompanying machine-readable sanity checks retain the C_D , C_L , and terminal-residual comparisons. At this mesh size, additional ranks reduce cell work but may increase halo synchronization; the rank-local diagnostics make that communication/load-balance tradeoff explicit.

8 Reproducibility, traceability, and sanity gates

The documented CMake command locates MPI, CGNS, and METIS through configurable dependency roots. The exact commands, ranks, measured time, step, and status appear in `run.manifest.csv`. Every report figure is mapped to its source CSV or PVTU file and named variable in `figure.manifest.csv`. `sanity_checks.json` records positive density/pressure, finite fields, symmetry, positive cylinder drag, unsteady lift, wall-condition semantics, skin-friction evidence, inviscid zero-viscous-force checks, figure identity, and rank consistency.

9 Limitations

The steady Rusanov baseline is deliberately robust but more dissipative than Roe, HLLC, or AUSM variants, especially at low Mach number and through the Mach-2 shock; transient HLLC still uses Rusanov for invalid star states. The current transport model is constant-viscosity laminar flow; turbulence, chemistry, general EOS, and 3-D elements remain extension interfaces rather than implemented models. The one-ring block-Jacobi solver trades simplicity and verifiable neighbour communication for more inner iterations than LU-SGS/Krylov, and rank-local restart currently requires the original MPI count. Cell-coloured VTK renderings preserve actual computed cells but do not perform higher-order visual interpolation. These limitations are not hidden in the convergence or status claims.

10 Conclusion

All required meshes, cases, output contracts, MPI demonstrations, physical sanity gates, and report artifacts completed from the submitted solver. The results support a maintainable baseline for higher-resolution fluxes, stronger implicit preconditioners, 3-D geometry, RANS, and general thermochemistry.